Data Analysis (AMS315)

Dr. Francis Ting

Stony Brook Institute at Anhui University

toting@ahu.edu.cn

September 8, 2025

Overview

1. Descriptive Statistics
2. Kurtosis
3. Skewness

Learning Objectives

In this chapter, you will learn:

- Basics of descriptive statistics
- Kurtosis
- Skewness

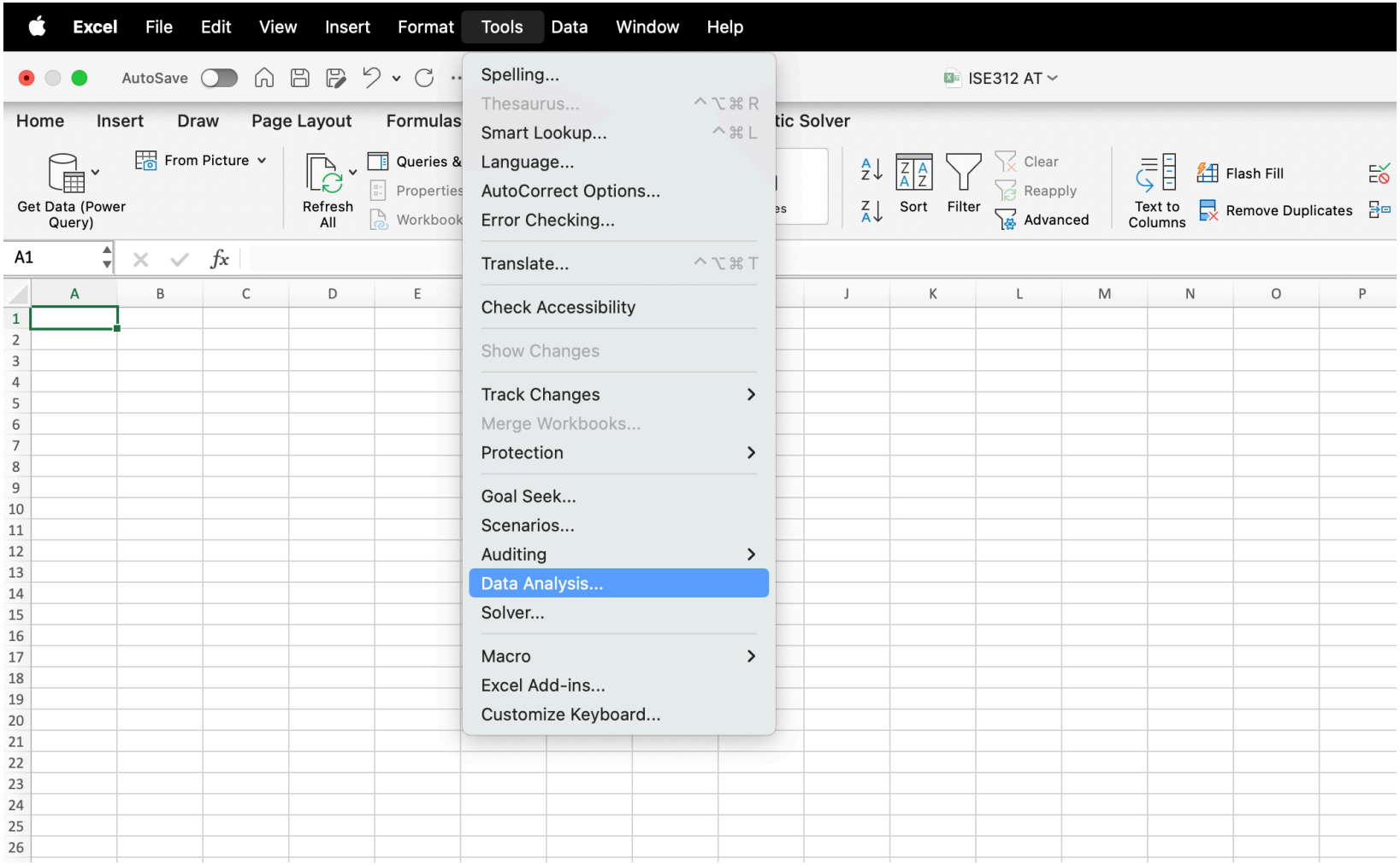
Descriptive Statistics

Descriptive statistics summarizes and describes the main features of a dataset. The main methods include:

- **Measures of Central Tendency** – Mean, median, and mode.
- **Measures of Dispersion** – Range, variance, standard deviation, **kurtosis**, and **skewness**.
- **Data Visualization** – Histograms, box plots, bar charts, and scatter plots.

Descriptive Statistics

Descriptive statistics for a dataset can be generated easily in MS Excel by clicking **Tools→ Data Analysis...**



Descriptive Statistics

Descriptive statistics for a dataset can be generated easily in MS Excel by clicking **Tools**→ **Data Analysis...**

	A	B
2		
3	Mean	76.712714
4	Standard Error	1.3472785
5	Median	78.94
6	Mode	90.725
7	Standard Deviation	13.805497
8	Sample Variance	190.59174
9	Kurtosis	2.1714711
10	Skewness	-1.4221679
11	Range	65.73
12	Minimum	29.37
13	Maximum	95.1
14	Sum	8054.835
15	Count	105
16	Confidence Level(95.0%)	2.6717038
17		

Kurtosis

Kurtosis is a statistical measure that describes the shape of a distribution's tails in relation to its overall shape, specifically focusing on the presence of outliers.

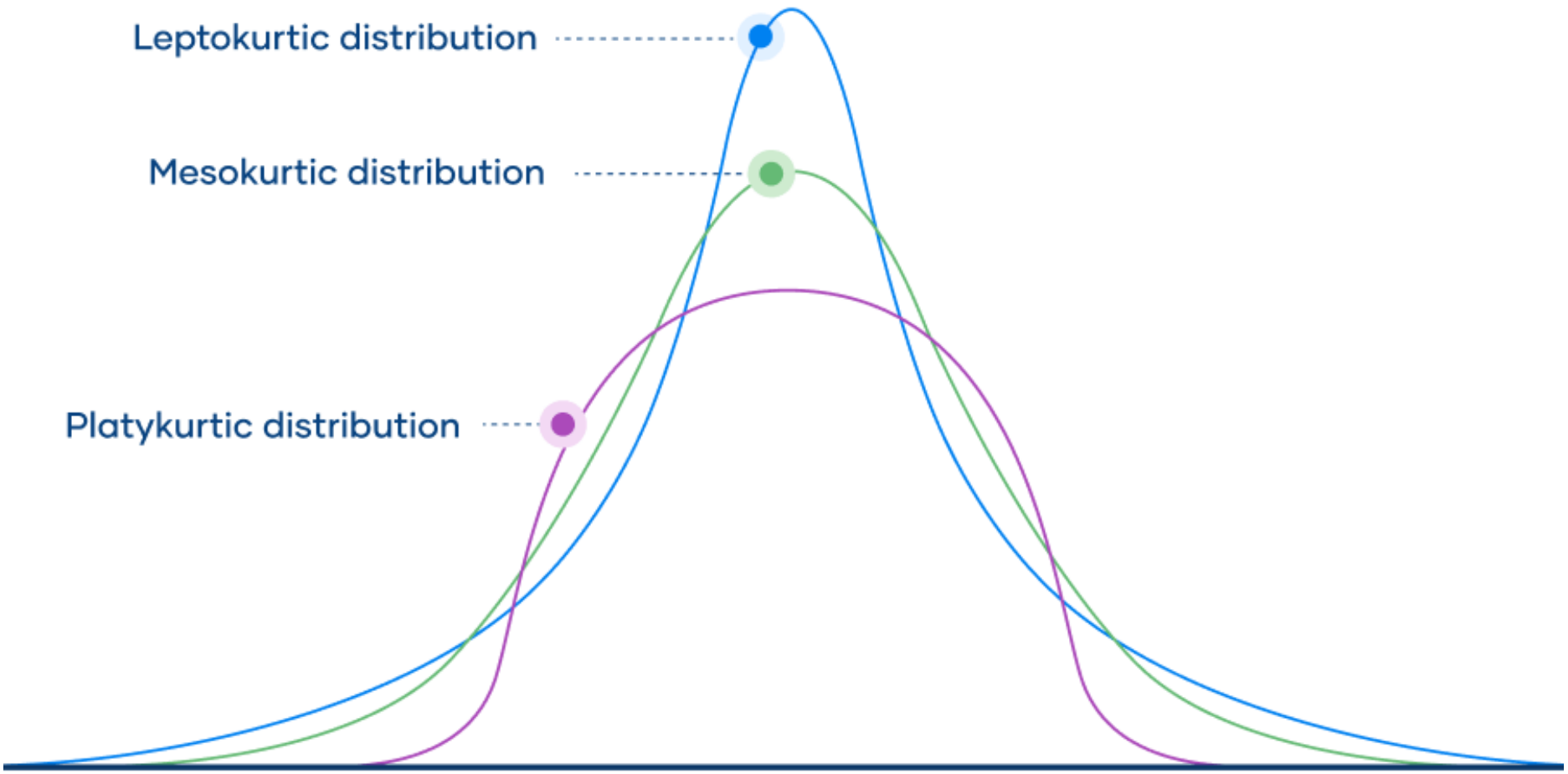
In other words, **kurtosis** tells us how much of the data is in the tails of the distribution, compared to a normal distribution.

Kurtosis

There are three types of Kurtosis:

1. **Mesokurtic** – A distribution with kurtosis (≈ 0). It has the distribution with kurtosis similar to that of a normal distribution.
2. **Leptokurtic** – A distribution with positive kurtosis (> 0). This indicates a distribution with heavy tails and a sharp peak. There are more extreme outliers, meaning more data points are found in the tails.
3. **Platykurtic** – A distribution with negative kurtosis (< 0). This indicates a distribution with light tails and a flatter peak. There are fewer extreme outliers, meaning fewer data points are found in the tails.

Kurtosis


Leptokurtic distribution
Mesokurtic distribution
Platykurtic distribution

Kurtosis

Here are the interpretations for Kurtosis distributions:

1. **Mesokurtic**– The bell curve has moderate tails and a moderate peak.
2. **Leptokurtic** – A distribution where more data points are concentrated around the mean, with fatter tails, indicating more outliers.
3. **Platykurtic** – A distribution that is flatter and has thinner tails, indicating fewer outliers.

Kurtosis

Kurtosis is calculated by:

$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

where:

$$\begin{aligned} n &= \text{The number of data points in the dataset,} \\ x_i &= \text{The } i\text{th data point in the dataset,} \\ \bar{x} &= \text{The mean of the data points,} \\ s &= \text{The standard deviation of the data points,} \end{aligned}$$

Kurtosis

Class Exercise:

1. Given an Excel spreadsheet consisting of a relevant dataset, calculate the Kurtosis value using the formula in the previous slide.
2. Verify the Kurtosis value by using **KURT** function in Excel.

Tips: Use **Shift+F4** to lock your cell when dragging the formula across cells.

Kurtosis

Usage in Data Analysis:

- **Kurtosis** is used alongside skewness to assess the normality of a distribution.
- While **skewness** measures the symmetry of the distribution, kurtosis measures the extremity of the distribution's tails.
- Understanding kurtosis helps in identifying the presence of **outliers** and the general shape of the data, which can be critical in certain statistical analyses and modeling tasks.

Skewness

- In descriptive statistics, **skewness** is a measure of the asymmetry or lack of symmetry in the distribution of data.
- It indicates the extent to which the distribution of data deviates from a perfectly **symmetrical**, normal distribution.

Skewness

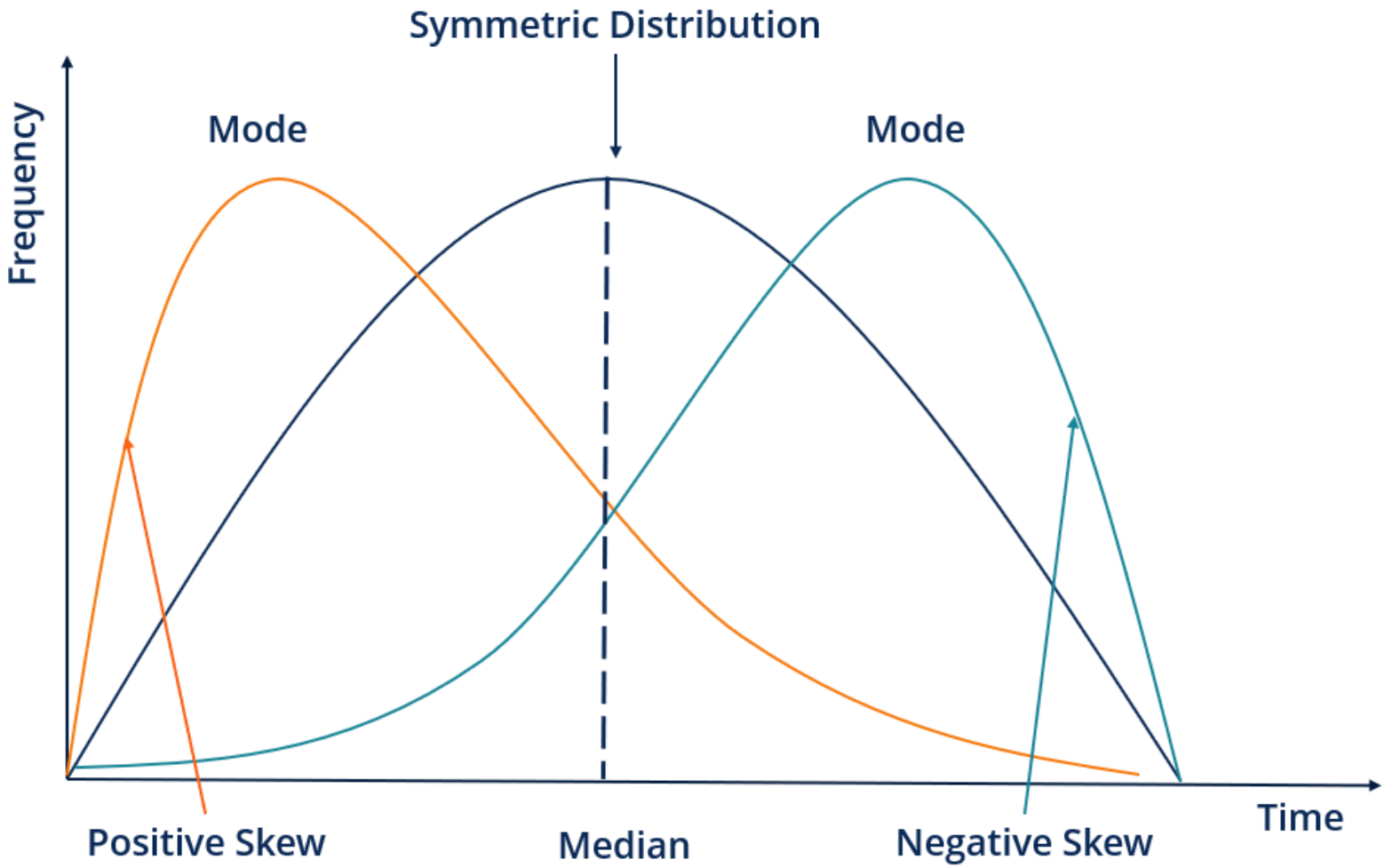
The formula to calculate **skewness** is:

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3$$

where:

$$\begin{aligned} n &= \text{The number of data points in the dataset,} \\ x_i &= \text{The } i\text{th data point in the dataset,} \\ \bar{x} &= \text{The mean of the data points,} \\ s &= \text{The standard deviation of the data points,} \end{aligned}$$

Skewness


Symmetric Distribution
Frequency
Mode
Mode
Positive Skew
Median
Negative Skew
Time

Kurtosis

The interpretations for **skewness** are:

1. **Right Skewness**(>0)− The right tail of the distribution is longer or fatter than the left tail. Most data points are concentrated on the **left** side of the distribution.
2. **Left Skewness** (<0)−The left tail of the distribution is longer or fatter than the right tail. Most data points are concentrated on the **right** side of the distribution.
3. **Zero Skewness** (=0) − The distribution is perfectly symmetrical. The tails on both sides of the mean are of equal length, resembling a **normal** distribution.

Skewness

Class Exercise:

1. Repeat the exercise above by calculating skewness of the same dataset.
2. Verify your answer by using the **SKEW** function in Excel.

Conclusions

At the end of this lesson, check that:

1. You can explain the basic concept of **descriptive statistics** to yourself and to your friend.
2. You understand the concept of **kurtosis** and know how to calculate the value in MS Excel.
3. You know the purpose of **skewness** and how the value is calculated in MS Excel.

The End